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Subject :-Introduction to Cyber Security

Unit -3 Social Media Overview and security

Contents

- Introduction social networks
- Social media :- Types, platforms, monitoring, hashtag, viral content, marketing, privacy, challenges, opportunities and pitfalls in online social networks
- Security issues related to social media
- Flagging and reporting of inappropriate content
- Laws regarding posting of in appropriate content
- Best practices for the use of social media

Case study: whatsapp spyware incident, Facebook-Cambridge Analytics data harvesting scandal

Social Network



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A social network is a structure made up of individuals or organizations (called nodes) that are connected by one or more types of relationships (called ties), such as friendship, family, work, shared interests, or information exchange.

-Family members, friends, coworkers, neighbourhood

Social Media Types and Platforms



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Types of Social Media and their respective platforms

- **1. Social Networking Sites**
- Platforms where users create profiles and connect with others.
  Examples: Facebook, LinkedIn
- **2. Media Sharing Networks**
- Used to share photos, videos, and other visual content.
  Examples: Instagram, YouTube, TikTok, Snapchat
- **3. Microblogging Platforms**
- Allow users to post short messages or updates.
  Examples: X (formerly Twitter), Threads
- **4. Messaging Apps**
- Enable private or group communication.

Types of Social Media and their respective platforms… continued

- **5. Blogging & Publishing Platforms**
- Used for longer articles and content publishing.
  Examples: WordPress, Medium, Blogger
- **6. Discussion Forums & Communities**
- Focus on discussions, questions, and knowledge sharing.
  Examples: Reddit, Quora
- **7. Social Bookmarking & Content Curation Sites**
- Users save, organize, and share online content.
  Examples: Pinterest

Types of Social Media and their respective platforms.. Continued.

- **8. Review Networks**
- Allow users to share opinions about products or services.
  Examples: Yelp, TripAdvisor
- **9. Live Streaming Platforms**
- Used for broadcasting live video content.
  Examples: Twitch, YouTube Live, Facebook Live
- In summary, social media platforms can be categorized based on how users interact—networking, sharing, messaging, publishing, reviewing, or streaming.

Social media monitoring



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Social media monitoring is the process of tracking, analyzing and responding to conversations, mentions and content shared about a brand, person ,organization or topic across social media platforms

Why Social Media Marketing is Important?



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Why Social Media Monitoring is Important

1.Brand Reputation Management

Helps identify positive and negative mentions to protect or improve brand image.

2.Customer Feedback & Support

Allows businesses to respond quickly to customer questions, complaints, or reviews.

3.Market Research

Provides insights into customer preferences, trends, and competitors.

4.Crisis Management

Detects potential issues early before they escalate.

5.Performance Tracking

Measures engagement, reach, and effectiveness of campaigns.

Hashtags

What is Monitored?
- Brand mentions
- Hashtags
- Comments and reviews
- Keywords related to products or services
- Competitor activity
- Trends and public opinion

Tools for Social Media Monitoring
- Hootsuite
- Sprout Social
- Brandwatch
- Google Alerts
- Buffer

Acitivity:
Search #kumbathon ,#nashik on google

Search google trends and analyze



A **hashtag** is a word or phrase preceded by the **# (hash) symbol** that is used on social media to identify and categorize content around a specific topic.

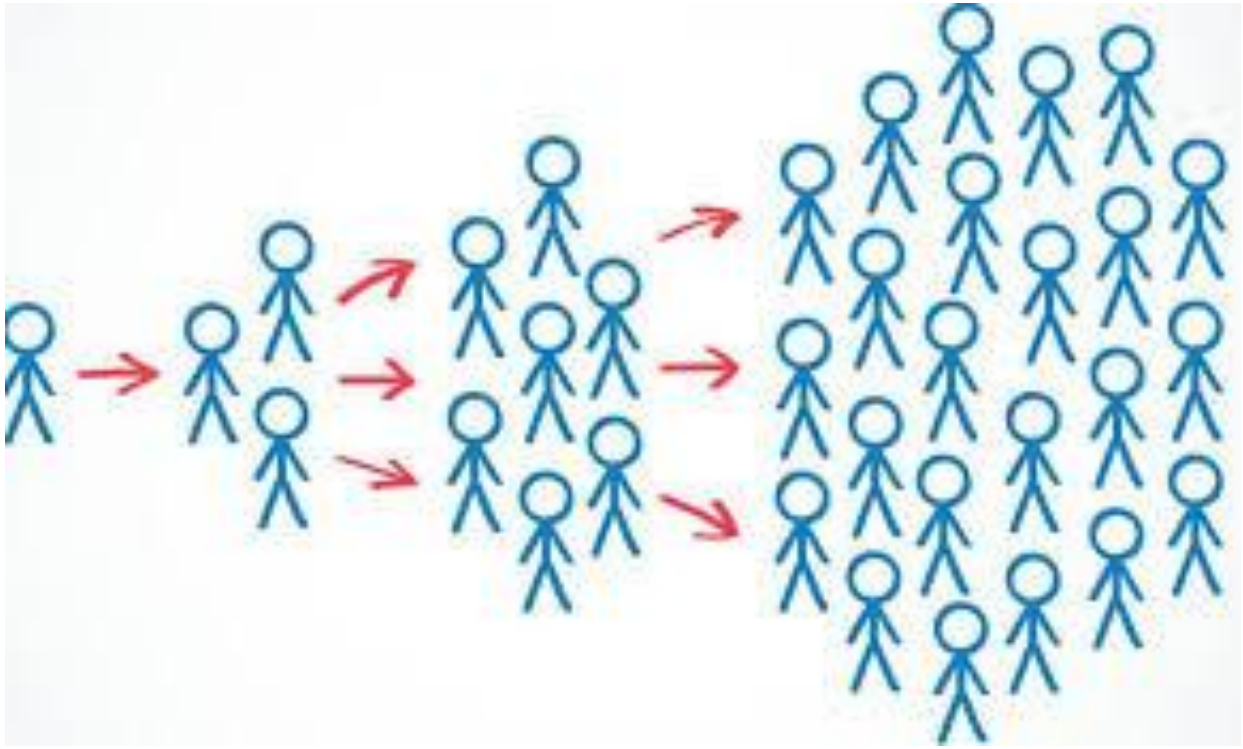
Sponsored Ads on Social media







Viral Content on social media



- **AI-style ChatGPT caricature images**
- — A new wave of 3D/AI-generated
- portrait trends is popping up across Instagran
- TikTok, and other platforms today.



Opportunities of Social Media

1.Global Reach & Connectivity

1. Connects people across countries instantly.
2. Builds communities and professional networks (e.g., LinkedIn, Facebook groups).

2.Marketing & Brand Promotion

1. Cost-effective way to advertise products/services.
2. Direct engagement with target audiences.

3.Customer Engagement & Feedback

1. Allows businesses to respond to queries, complaints, or reviews in real-time.
2. Collects valuable data on customer preferences.

4.Information & Awareness

1. Spreads news, educational content, and awareness campaigns quickly.
2. Enables social movements and advocacy.

5. **Creativity & Content Sharing**

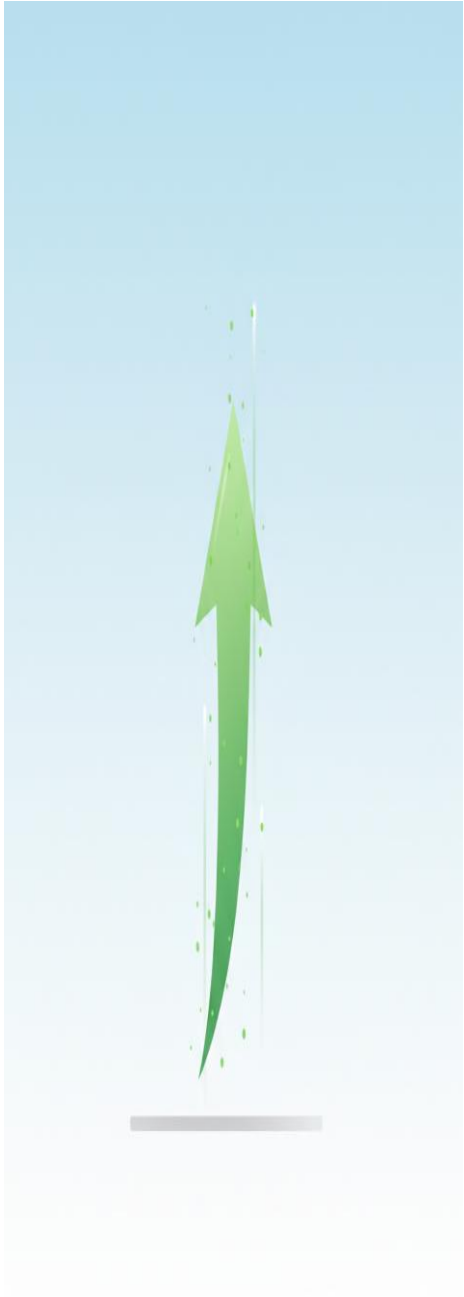
•Platforms like TikTok, Instagram, and YouTube encourage creative expression.

•Opportunities for influencers, artists, and small businesses to grow.

6. Learning & Professional Development

•Online communities provide tutorials, courses, and knowledge-sharing.

•Networking opportunities for careers and collaborations.



Challenges and Pitfalls in Social Media



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- **Privacy & Security Risks**
- Data breaches and identity theft.
- Oversharing personal information can be dangerous.
- **Misinformation & Fake News**
- False information spreads quickly.
- Difficult for users to verify credibility.
- **Cyberbullying & Online Harassment**
- Harassment, trolling, or negative comments affect mental health.
- **Addiction & Mental Health Issues**
- Excessive use can lead to anxiety, stress, or reduced productivity.
- Fear of missing out (FOMO) is common.
- **Reputation Management**
- Negative posts or viral controversies can harm individuals or brands.
- **Algorithm Bias & Echo Chambers**
- Social media may create filter bubbles, limiting exposure to diverse views.
- Algorithms can unintentionally amplify polarizing content.

Challenges and Pitfalls in Social Media

8. Scams & Fraud

•Phishing, fake giveaways, or investment scams.

•Impersonation and identity theft risks.

9. Over-commercialization

•Excessive ads or sponsored content reducing user trust.

•Can make platforms feel less authentic or manipulative.





Just an example of commercialization no harm or opinion on stardom

Security Issues in Social Media

1. Privacy Breaches
•Personal data (like photos, location, or contact details) being exposed or misused.
•Platforms sometimes collect more information than users realize.
2. Account Hacking
•Cybercriminals can gain access to accounts through weak passwords or phishing.
•Leads to impersonation, fraud, or spreading malware.
3. Phishing Attacks
•Fake messages or links trick users into revealing passwords or financial information.
•Can appear as friend requests, direct messages, or fake platform notifications.
4. Malware & Viruses
•Malicious links, files, or ads shared through social media can infect devices.
•Some malware spreads automatically through social connections.
5. Identity Theft
Hackers can steal personal information to create fake accounts or commit fraud.
Can damage reputation or lead to financial loss.
6. Cyberstalking & Harassment
Unauthorized tracking of users' activities or location.
Repeated harassment can threaten personal safety.
7. Data Mining & Misuse
Third-party apps may access user data without consent.
Data can be sold for targeted advertising or political campaigns.
8. Weak Security Settings
Users often leave accounts public or use weak passwords.
Default settings may expose more information than intended.

Flagging of inappropriate content

- Hate speech or harassment
- Bullying or threats of violence
- Graphic violence
- Scams or fraud
- Impersonation
- Misinformation that could cause harm
- Non-consensual intimate images
- Spam

Reporting of inappropriate content

- **Find the three dots (⋯) or options menu** near the post, comment, or profile.
- Click **“Report”** or **“Flag.”**
- Choose the reason that best fits.
- Submit the report.
- (Optional) Block or mute the account.



What to do when?

- **Block** → If someone is bothering you personally.
- **Report** → If the content violates platform rules.
- **Do both** → If harassment or threats are involved.

Laws regarding posting of in appropriate content

Defamation

Cyber bulling

Hate speech

Sensitive content without concern

Copyright infringement

Privacy violation

Platform penalities



Best practices for the use of social media

- Do not try to comment your opinion on social topic which can harm individual or any element in society

Create Valuable, Shareable Content

Content that performs usually does at least one of these:

Educates

Entertains

Inspires

Solves a specific problem

Sparks emotion or discussion



Best practices:(Reply)

Reply to every comment

Ask questions in posts

Use polls and interactive features

Engage with others in your niche daily

Tip: Spend 15–20 minutes engaging before and after posting.

Avoid Common Mistakes

•Posting without a goal

•Inconsistent branding

•Overusing hashtags

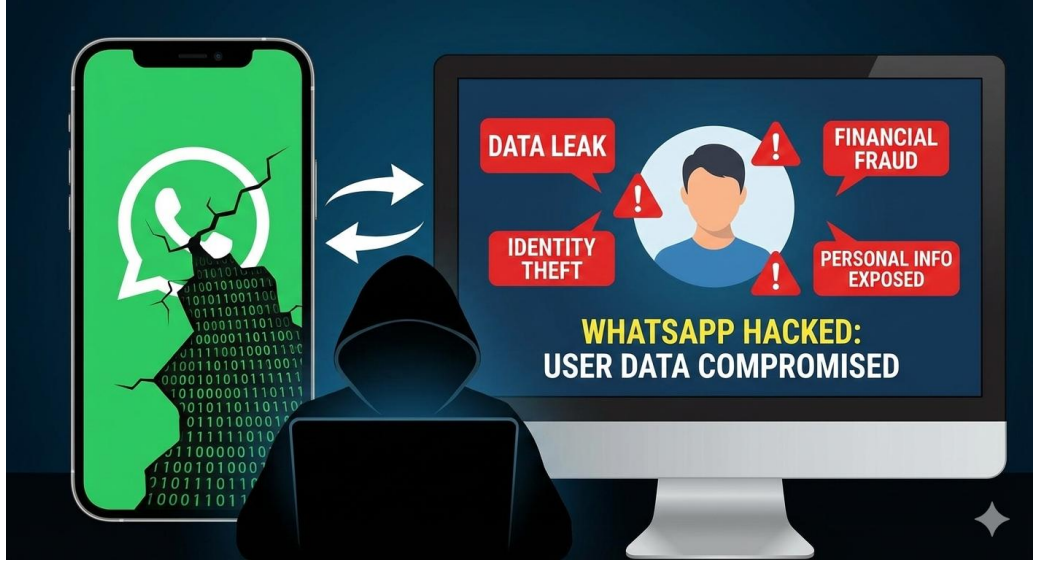
•Being overly promotional

•Ignoring comments

•Copying competitors blindly

Think Long Term

Case study: whatsapp spyware incident



In **May 2019**, WhatsApp discovered that a highly sophisticated spyware tool called **Pegasus** — developed by Israeli company **NSO Group** — was being used to infect users' phones **without their knowledge** by exploiting a vulnerability in WhatsApp's voice-calling feature.

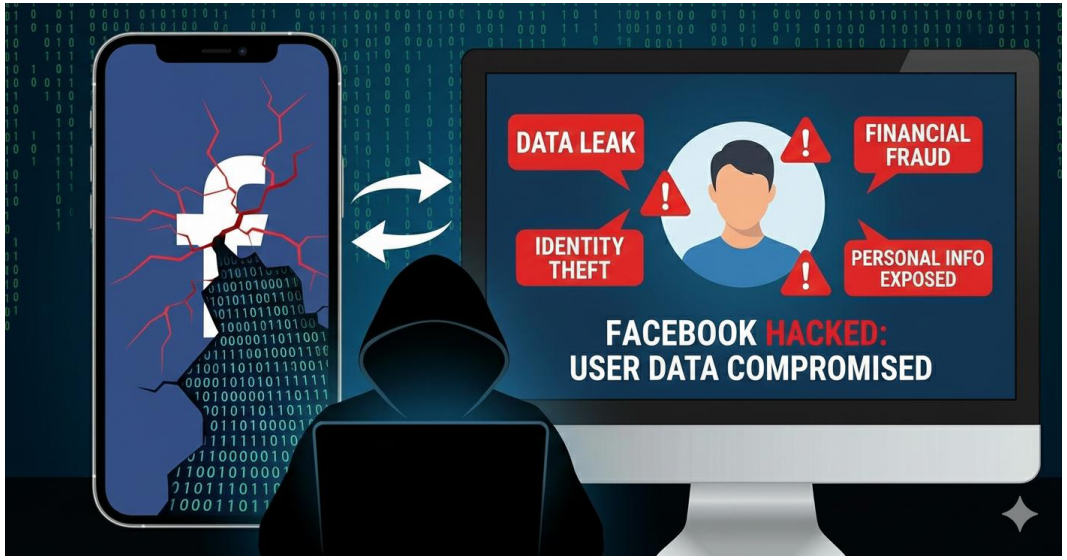
Around **1,400 WhatsApp users in 51 countries were targeted**, including *human rights defenders, journalists, activists, diplomats, and officials*.

WhatsApp, owned by **Meta**, filed a lawsuit in the U.S. in **October 2019**, accusing NSO Group of:
Hacking into WhatsApp servers without authorization,

U.S. federal jury ordered NSO Group to **pay roughly $167.25 million (~₹1,400 crore)** in punitive damages and about **$444,719 in compensatory damages** to Meta/WhatsApp.

Case study: Facebook-Cambridge Analytics data harvesting scandal

Between **2013–2015**, a political consulting firm called **Cambridge Analytica** obtained personal data from Facebook users without their proper consent.



How the Data Was Collected

1.A researcher, **Aleksandr Kogan**, created a Facebook personality quiz app called:
"This Is Your Digital Life"

2.About **270,000 users** installed the app and consented to share their data.

3.At the time, Facebook's platform rules allowed the app to also collect data from users' **friends** — without those friends' consent.

4.As a result, data from approximately **87 million Facebook users** was harvested.

5.The data was later transferred to **Cambridge Analytica**, which violated Facebook's policies.

THANK YOU